Data Quality Audit Report

File: audit_20260510_170601_640783_Titanic-Dataset.csv
Rows: 891
Columns: 12
Audit date/time: 2026-05-10 17:06:12

Section A — Summary of Findings

Column Name	Issue Type	Affected Records	% of Total
Age	Missing values	177	19.87%
Cabin	Missing values	687	77.10%
Embarked	Missing values	2	0.22%
Cabin	Null concentration	687	77.10%
Ticket	Mixed data types	230	25.81%
Age	Impossible / out-of-range values	7	0.79%
Ticket	Numeric values stored as text	661	74.19%
Ticket	Inconsistent casing	667	74.86%
Name	Leading / trailing whitespace	2	0.22%
Age	Statistical outliers	11	1.23%
SibSp	Statistical outliers	46	5.16%
Fare	Statistical outliers	116	13.02%

Section B — Detailed Narrative

The column 'Age' had 177 instances where values did not exist, indicating incomplete provision of information. Additionally, 7 instances were identified where Age indicated values of 0.83 or less, or 0.92 or less, which violates the expected age range of greater than 0 and less than or equal to 120. Furthermore, 11 instances of Age were flagged as outliers, with values such as 66.0, 65.0, and 71.0 noted.

The column 'Cabin' presented a significant proportion of missing values, with 687 instances, representing 77.1% of the total dataset. This high concentration of null values, also indicated by the null concentration check, suggests a substantial lack of information for this field.

The column 'Embarked' contained 2 instances of missing values, which is 0.22% of the total dataset, indicating a minor incompleteness in the provided information.

The column 'Ticket' exhibited mixed data types, with 230 instances, or 25.81% of the dataset, being of string type, while the majority type was integer. A further 661 instances, representing 74.19% of the data, were identified as numeric values stored as text. Casing inconsistencies were also noted in 667 instances, or 74.86% of the dataset.

The column 'Name' showed 2 instances with whitespace issues, such as trailing spaces, which is 0.22% of the dataset.

The column 'SibSp' identified 46 instances of outliers, representing 5.16% of the dataset, with values like 3, 4, and 5 noted.

The column 'Fare' presented 116 instances of outliers, accounting for 13.02% of the dataset, including values such as 71.2833 and 263.0.

No dataset-level issues were identified.